Module-8

Hologram based Structural Obfuscation for DSP Cores

Dr Anirban Sengupta

Associate Professor, CSE,

IIT Indore-India

Outline

- ❑ Introduction
- ❑ Background on Image Hologram
- ❑ Applying Image Hologram concept for Structural Obfuscation
- ❑ Hologram based Obfuscation Methodology
- ❑ Illustrative examples of Hologram based Obfuscation for DSP cores
- ❑ Determination of Gate count affected due to Hologram based Obfuscation of DSP cores
- ❑ Analysis on Case Studies

Introduction

- A digital signal processing (DSP) kernel inside a digital signal processor executes DSP algorithms to facilitate specific applications such as
    - de-noising, compression-decompression, correlation and transformation etc.
- Owing to huge applications of DSP kernels in modern smart electronic devices, their security perspective has become very important.
- The structural obfuscation is one of the major security mechanisms that can be employed to secure the DSP designs against reverse engineering threats resulting into attacks such as Trojan insertion, piracy etc.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Background on Image Hologram

- Image hologram is special feature whereby multiple images can be overlapped with each other, while simultaneously being capable for display at the same time.
- The overlapping process is similar to concept of camouflaging.
- Different images can be viewed at different angles in an image hologram. Different angles of viewing can be as follows: horizontal to vertical (in both directions) or upside to bottom (in both directions).
- At different angles of viewing, distinct images evolve through switch elements (Hologram Features, Matrix Technologies).
- This phenomenon can be a useful tool for performing structural obfuscation of DSP circuits as well.

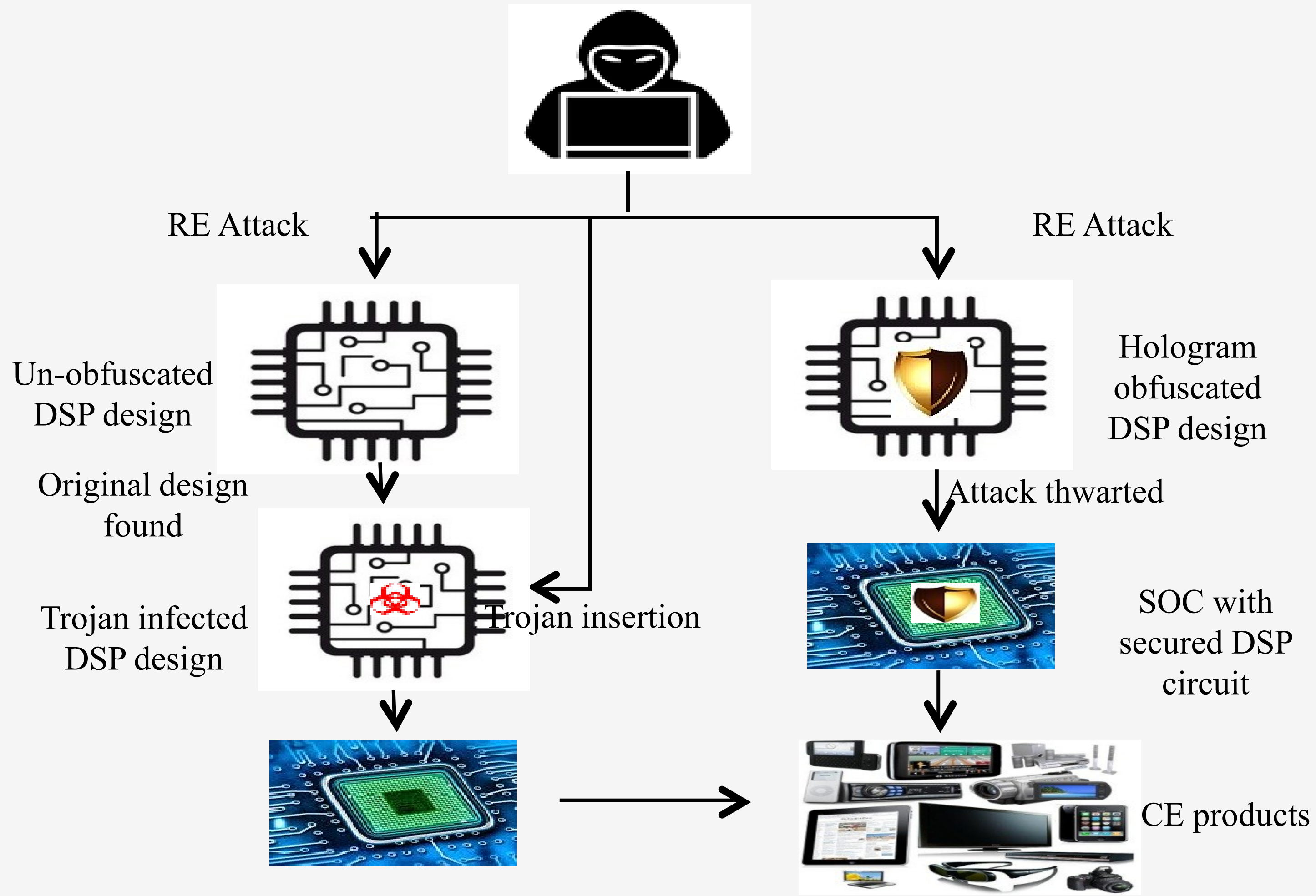
Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Applying Image Hologram concept for Structural Obfuscation

- Similar to image hologram concept, hologram based obfuscation has been formulated such that atleast two DSP applications are integrated together in a camouflaged fashion.
- These camouflaged DSP applications (in circuit form) are possible to be viewed and activated at specific switching value.
- In other words, individual DSP circuit architecture is functional based on the application of specific bit combination.
- Similar to image hologram, these bit patterns are equivalent to the switching element used for viewing different images.
- Introduced a hologram based obfuscation methodology where the concept of camouflaging is inspired from image hologram phenomenon.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

A conceptual view of protecting DSP designs using hologram obfuscation



Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

- Hologram based obfuscation designs provide the necessary camouflaging to thwart reverse engineering (RE) attacks as it becomes un-obvious (post obfuscation) for an attacker to identify the architecture of the application.
- Hologram based obfuscated designs enhance the complexity of understanding the functionality of the architecture, thus hindering RE attacks.
- One of the most important aspects of an obfuscation approach is to perform obfuscation at minimal design overhead.
- Hologram based obfuscation satisfies this property as it obfuscates two applications simultaneously into a single design architecture.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- **Features of Hologram based obfuscation**
  - Performs structural obfuscation during datapath and controller generation phases of high level synthesis (HLS) for DSP applications.
  - Utilizes image hologram camouflaging concept for structural obfuscation.
  - Integration of two DSP applications into a single design architecture using multiple obfuscation algorithms.
  - Capable to achieve massive savings in utilization of resources owing to obfuscating two applications simultaneously compared to two individually obfuscated designs.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- **Threat Model**
  - Hologram based obfuscation is capable of thwarting reverse engineering (RE) attacks that are used to implant malicious logics into the design, perform cloning/piracy or fraud claim of ownership.
  - This form of structural obfuscation enhances the complexity of launching RE attacks by hindering an attacker from identifying the actual functional architecture of the design.
  - In other words, it confuses an adversary by making the design architecture un-obvious.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- ## Overview

Input

Scheduled Data Flow Graph (DFG) of two DSP applications

Perform hologram based obfuscation during high level synthesis (HLS)

Rule #1: Multiplexing of inputs of only similar portions of both applications

Rule #2: Multiplexing of all inputs of both applications

Rule #3: Multiplexing of inputs of only similar components of both applications

Hologram obfuscated RTL design

Output

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

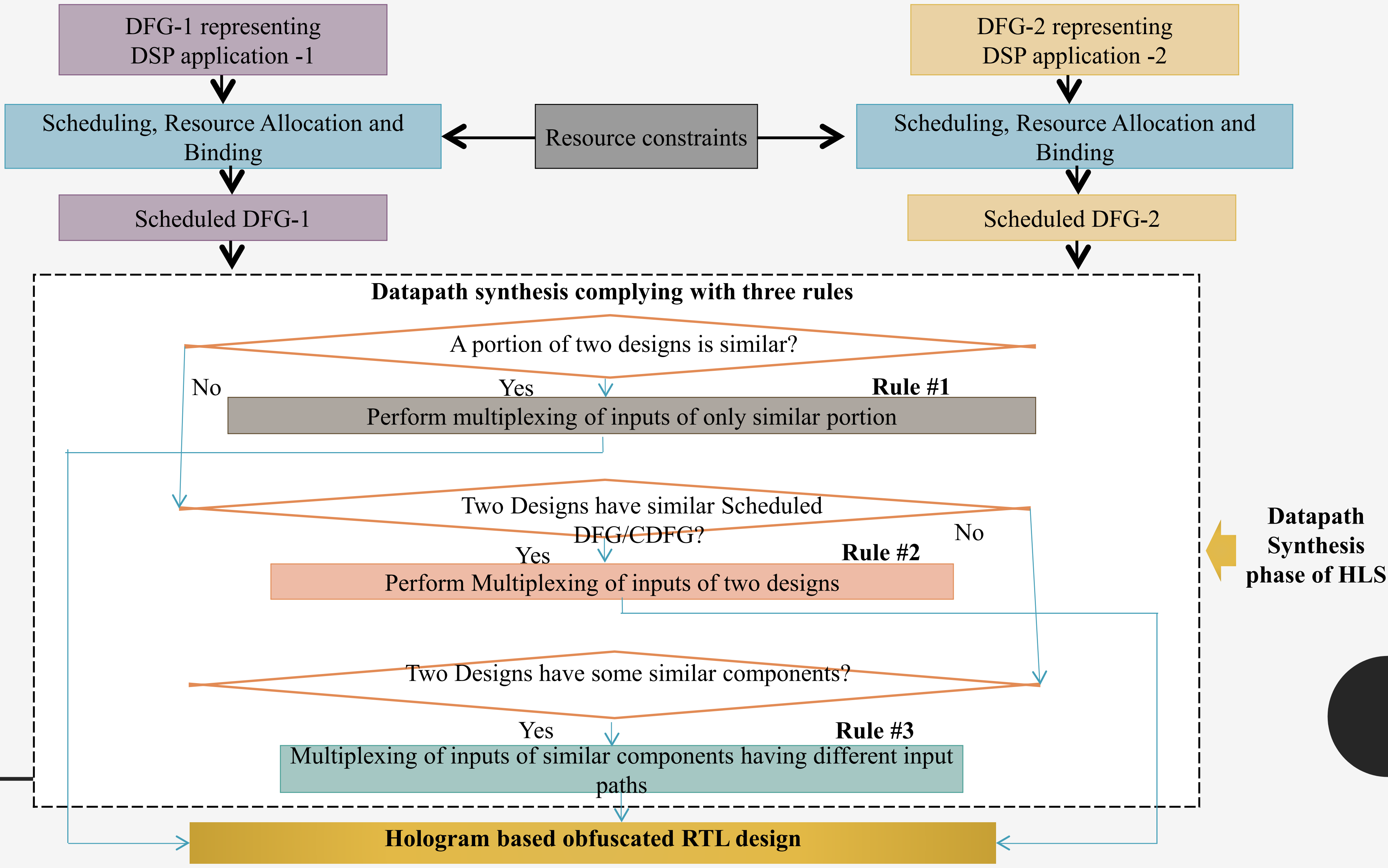
Hologram based Obfuscation Methodology

- **Overview**
  - The inputs to this approach are scheduled DFG of two DSP applications and output is a hologram obfuscated design that is a camouflaged integrated RTL datapath.
  - The hologram based obfuscation is performed in HLS framework using three rules.
  - According to the first rule, only a sub-set of inputs of both DSP applications are multiplexed. According to the second rule, all inputs of both DSP applications are multiplexed; while according to the third rule, only inputs of similar components of both DSP applications are multiplexed.
  - Depending on the design architecture of two selected DSP applications, the aforementioned rules are applied for generating a hologram obfuscated design

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- **Details**



Hologram based Obfuscation Methodology

- *Rule #1:Multiplexing sub-set of inputs of two DSP applications:*
  - If a portion of scheduled DFGs of two different DSP applications is structurally identical (identical portions of both applications require similar resources with similar input-output connectivity), then following two actions are performed:

➢ The inputs of only similar portions of both applications are multiplexed.

➢ The outputs of both applications are also multiplexed.

- The switching between two designs is performed using multiplexers (Muxes) each of size 2:1. At select line 'H'=0, one DSP design activates (gets inputs through switching Muxes) while at 'H'=1, another DSP design activates.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- *Rule #2:Multiplexing of all inputs of two DSP applications:*

  - If two scheduled DFGs representing two different DSP applications are fully identical (i.e. both applications require same number of inputs and outputs and similar resources with similar input-output connectivity), then following action is performed:
  - All inputs of both applications are multiplexed.
  - At ‘H’=0, one DSP design gets functionally enabled while at ‘H’=1, functionality of another DSP design gets enabled.

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Hologram based Obfuscation Methodology

- ❖ *Rule #3:Multiplexing of inputs of similar components in two DSP applications:*
  - ➢ This rule is applied when two applications have some similar components in their datapaths; however the paths of inputs of those components are different.
  - ➢ When this condition is satisfied, multiplexing of inputs of only similar components of both applications is performed.
  - ➢ Components of one application accept inputs at ‘H’=0while components of the other application accept inputs at ‘H’=1.
- ❑ When these rules are applied on scheduled DFGs of two DSP applications and processed through a HLS framework, then a hologram obfuscated RTL design is generated

Illustrative examples of Hologram based Obfuscation for DSP cores

- ***Case study of IIR-FIR filter obfuscated design***

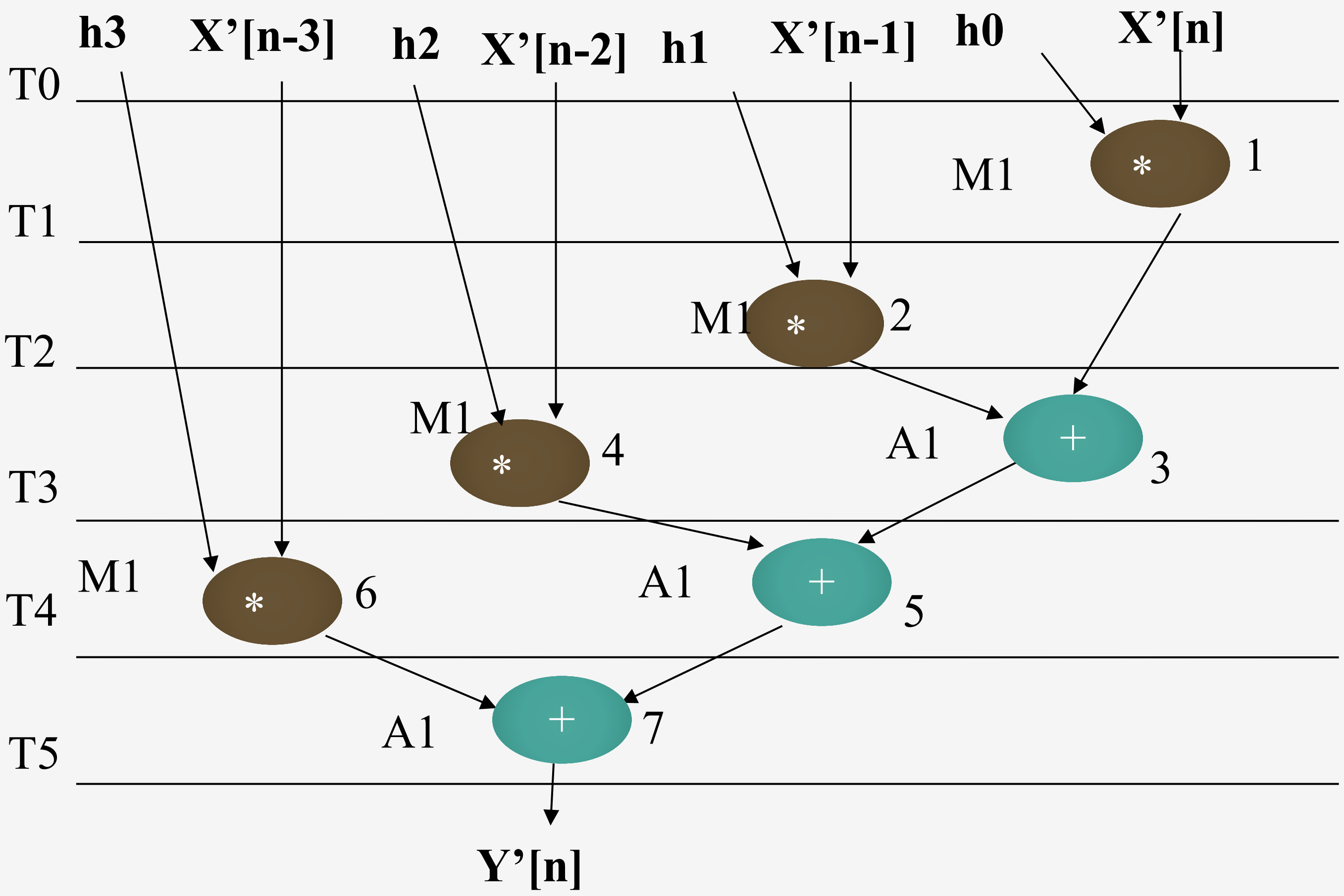
- The generic equation of IIR and FIR filters are as follows:
- **IIR:** Y[n]=bo*X[n]+ b1*X[n-1]+ b2*X[n-2]+b3*X[n-3]- a1*Y[n-1]- a2*Y[n-2]-a3*Y[n-3]
- **FIR:** Y’[n]=ho*X’[n]+ h1*X’[n-1]+ h2*X’[n-2]+ h3*X’[n-3]
- Where, Y[n] and Y’[n] represent the output of IIR and FIR filters respectively. Further,
- {a1, a2, a3, bo, b1, b2, b3}=Set of input coefficients of IIR filter.
- {ho, h1, h2, h3}=Set of input coefficients of FIR filter.
- {X[n], X[n-1], X[n-2], X[n-3]}=Set of previous values of input of IIR filter.
- {Y[n-1], Y[n-2], Y[n-3]}= Set of previous values of output of IIR filter.
- {X’[n-1], X’[n-2], X’[n-3]}=Set of previous values of input of FIR filter.

❖ *Case study of IIR-FIR filter obfuscated design*

- The first step is to create DFG of both DSP applications.
- Further, the scheduling is performed on both DFGs based on resource (functional unit) constraints of one adder and one multiplier (1A, 1M).
- Once the scheduled DFG of IIR and FIR filters are obtained, they are fed to datapath synthesis phase of HLS in order to generate a common obfuscated RTL of both DSP applications.
- Here, hologram based obfuscation is employed using rule #1. This is because structurally a portion of IIR filter is identical to FIR filter.

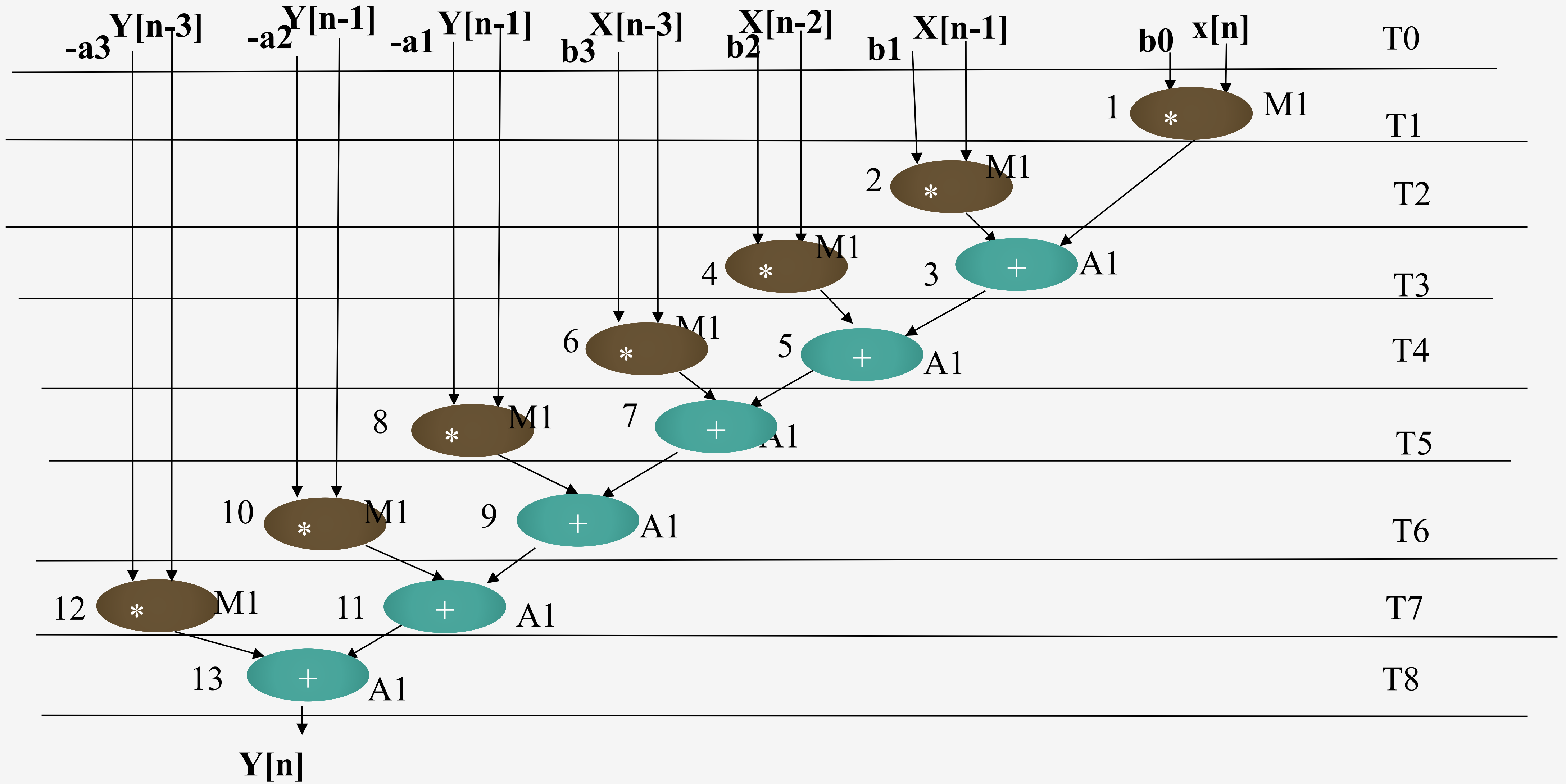
Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

❖ *Case study of IIR-FIR filter obfuscated design*



Scheduled DFG of FIR filter based on 1 adder and 1multiplier

❖ *Case study of IIR-FIR filter obfuscated design*



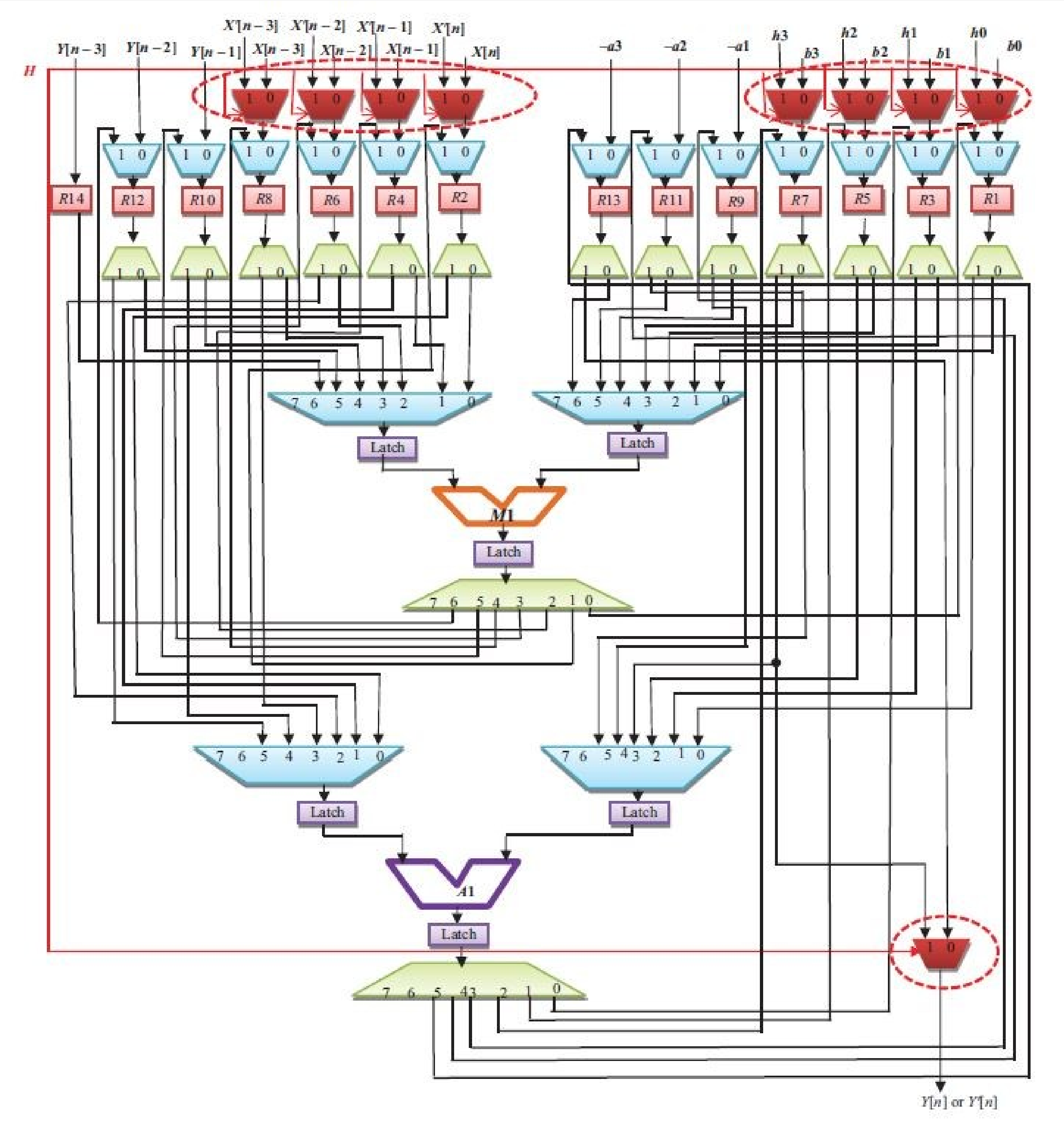
Scheduling of IIR filter based on resource constraints (1A, 1M)

❖ *Case study of IIR-FIR filter obfuscated design*

- As per rule #1, inputs of only similar portions of both designs are multiplexed through a number of 2:1 Muxes and output of both design are also multiplexed through a single 2:1 Mux.
- Based on the value of select line ‘H’, either of the design gets activated.
- Application of the specific bit combination in the hologram design, results into manifestation of the respective datapath architecture.
- For example, at ‘H’=0, IIR design gets functionally enabled while at ‘H’=1, FIR design gets activated. Here, the switching element of the hologram obfuscated design is the select control of the 2:1 Muxes which can be directly controlled by the IP vendor/designer.

❖ Case study of IIR-FIR filter obfuscated design

- *Hologram obfuscated RTL datapath of IIR-FIR filter integrated design*
- *Note: switching Muxes of hologram have been encircled in red coloured dotted ovals.*



- *Resemblance of an image hologram with a hologram obfuscated design*
  - Switching or flipping elements integrated in an image hologram are responsible for switching between two (or more) images at different viewing angles.
  - Likewise in a hologram obfuscated design, a number of additional multiplexers (acting as switching elements) have been used for mimicking hologram feature.
  - The switching Muxes of a hologram obfuscated design have been shown with red coloured Muxes in the integrated RTL datapath (select/control lines of switching Muxes have also been shown with red colour).

Illustrative examples of Hologram based Obfuscation for DSP cores

❖ *Case study of 4-point DCT-IDCT obfuscated design*

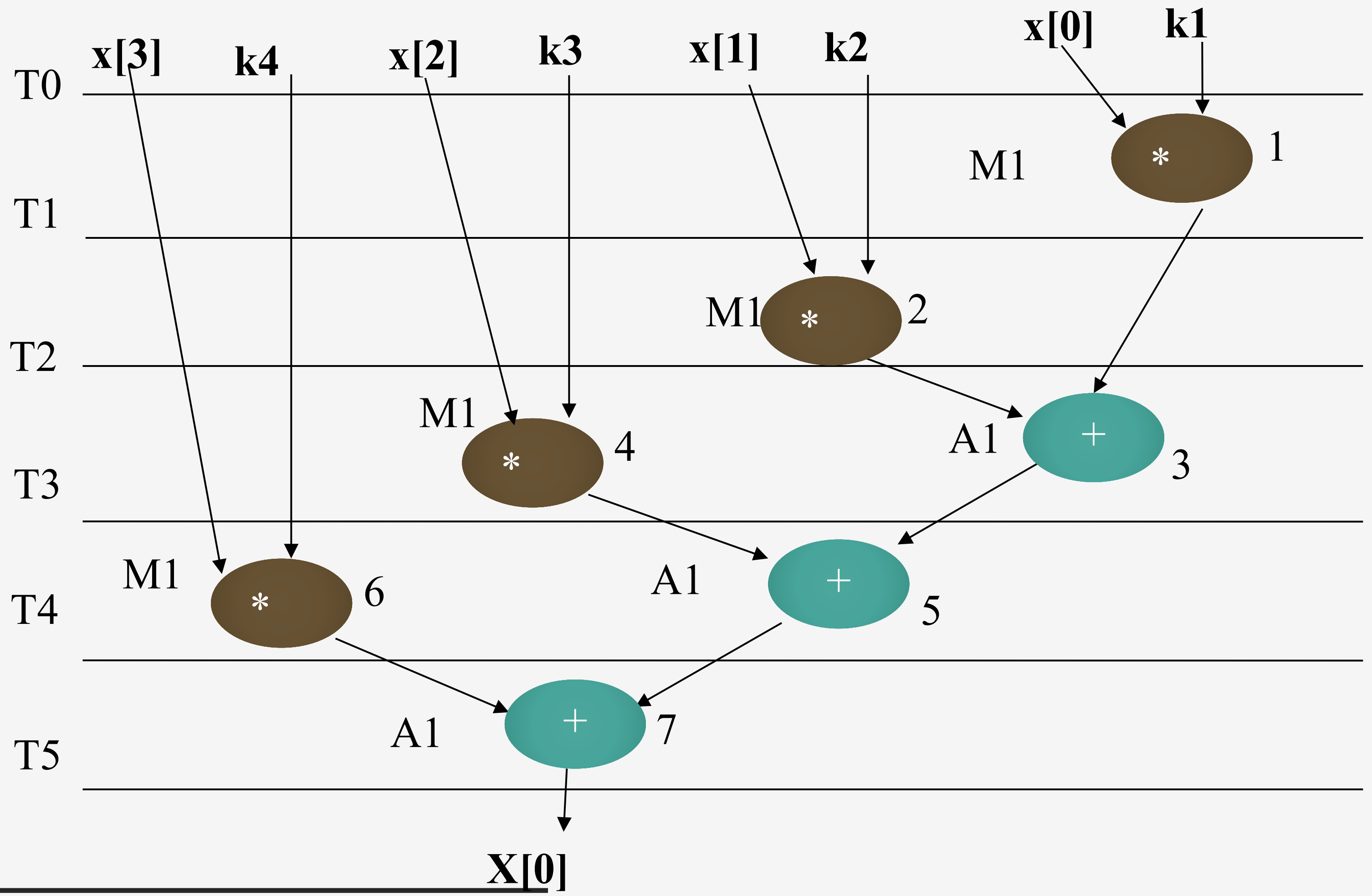
- The generic equation of 4-point DCT and 4-point IDCT are as follows :
- **4-Point DCT**: X[0]= k1*x[0]+ k2*x[1]+ k3*x[2]+ k4*x[3]
- **4-Point IDCT:** x'[0]= k'1*X'[0]+ k'2*X'[1]+ k'3*X'[2]+k'4*X'[3](4)

- Where, X[0] and x'[0] represents the first output value of 4-point DCT and 4-point IDCT respectively. Further,
- {k1, k2, k3,k4}= Set of generic values of input coefficients of 4-point DCT.
- {k'1, k'2, k'3,k'4}= Set of generic values of input coefficients of 4-point IDCT.
- {x[0], x[1], x[2], x[3]}= Set of input values of 4-point DCT.
- {X'[0], X'[1], X'[2], X'[3]}= Set of input values of 4-point IDCT.

❖ *Case study of 4-point DCT-IDCT obfuscated design*

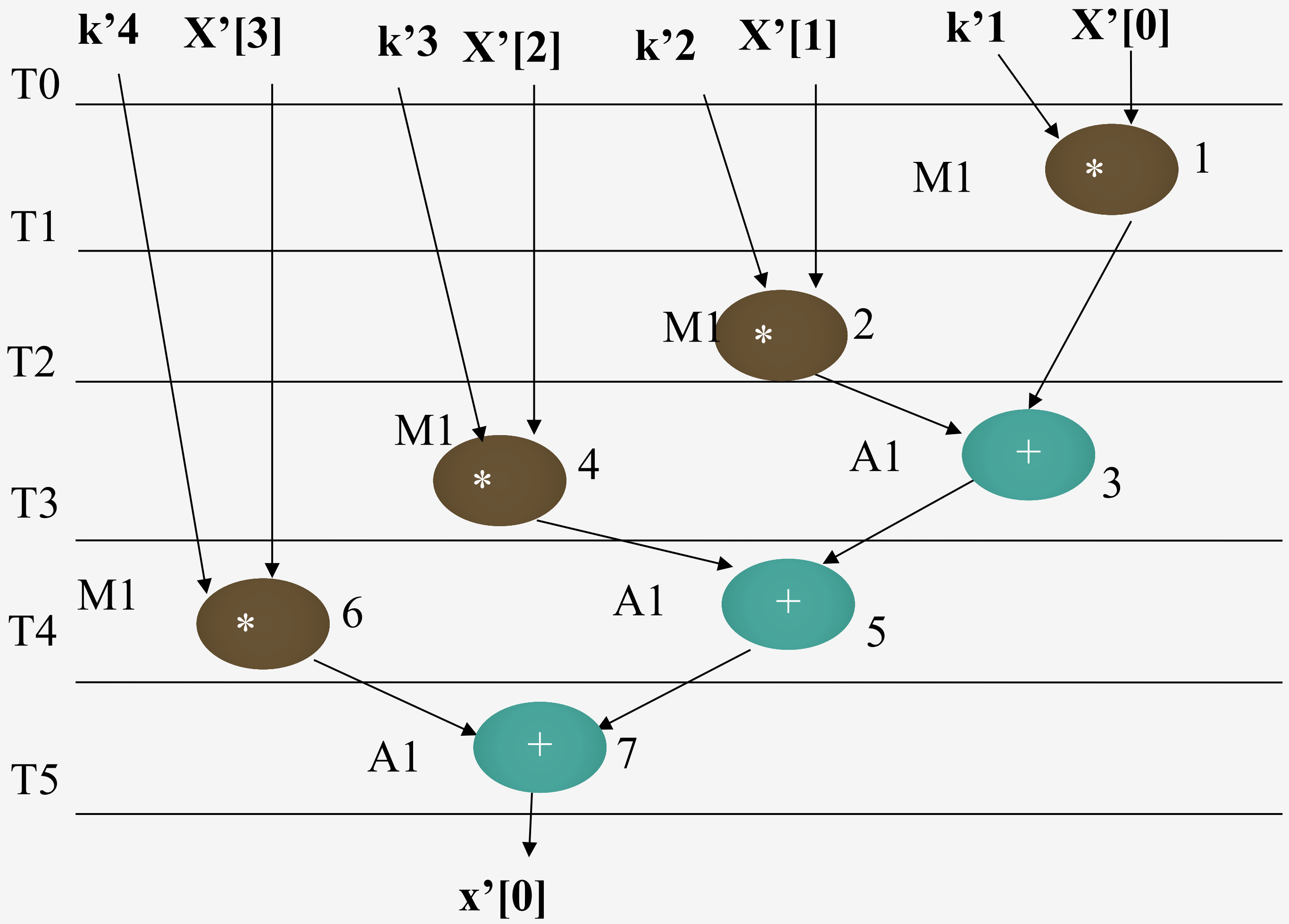
- The first step is to create DFG of both DSP applications.
- Further, the scheduled DFGs of 4-point DCT and 4-point IDCT (based on resource constraints of 1A, 1M) are obtained.
- Further, they are fed to datapath synthesis phase of HLS in order to generate a common obfuscated RTL of both DSP applications.
- Here, hologram based obfuscation is employed based on obfuscation rule #2.
- This is because structurally a 4-point DCT is identical to 4-point IDCT

❖ Case study of 4-point DCT-IDCT obfuscated design



Scheduled DFG of 4-point DCT based on 1 adder and 1 multiplier

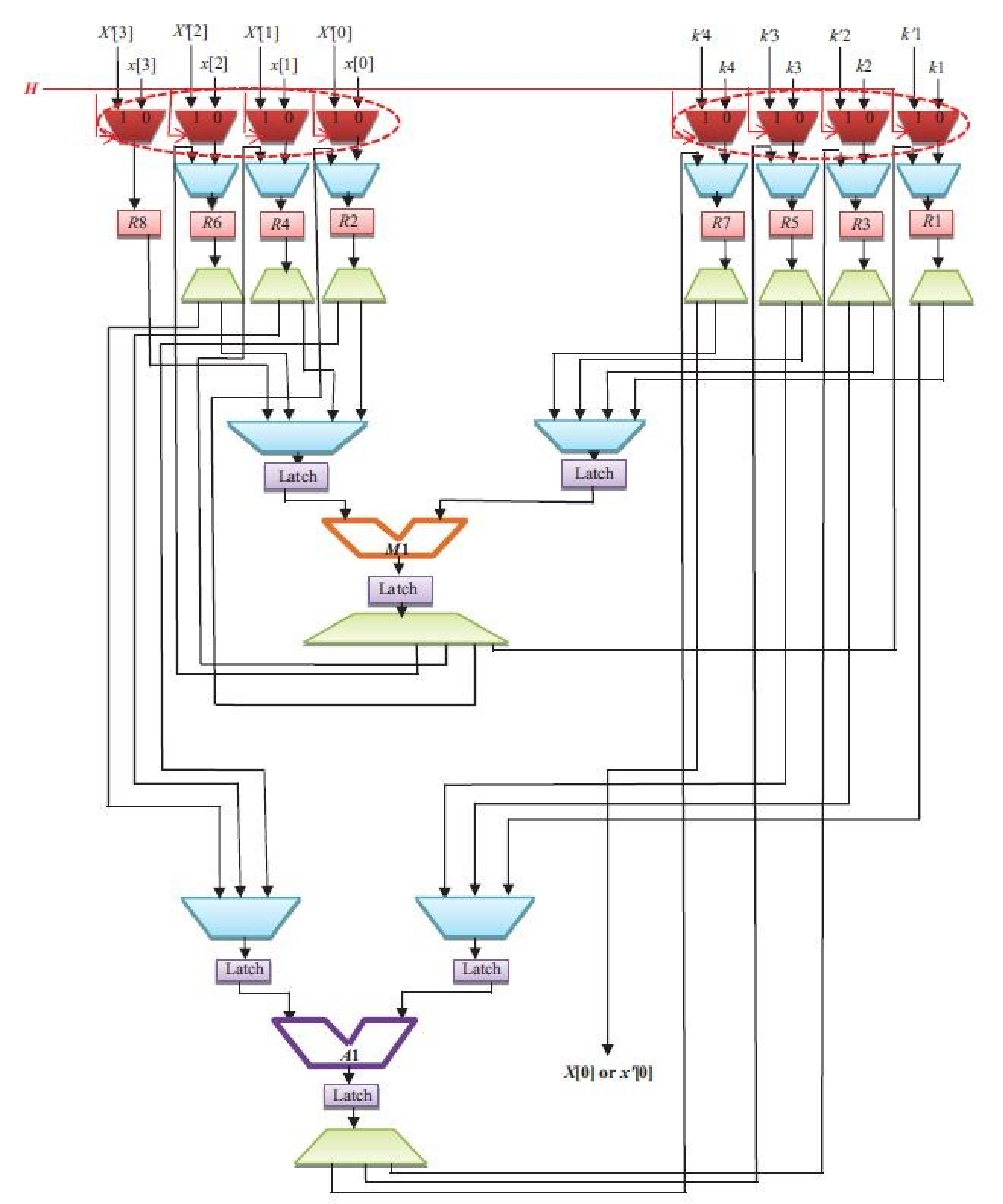
❖ Case study of 4-point DCT-IDCT obfuscated design



Scheduled DFG of 4-point IDCT based on 1 adder and 1multiplier

❖ *Case study of 4-point DCT-IDCT obfuscated design*

- ➢ *Hologram obfuscated RTL datapath of 4-pont DCT-IDCT integrated design*



Determination of Gate count affected due to Hologram based Obfuscation of DSP cores

❖ The process of estimating gate count of both hologram obfuscated design and HLT obfuscated design can be generically represented using the following equation:

$$G_c^a = \Delta G_c^b + G_c^i$$

- Where, G_c^a = Number of gates affected due to obfuscation.
- ΔG_c^b = Difference in gate count between obfuscated design and baseline.
- G_c^i = Number of gates modified in terms of input connectivity after obfuscation.

➢ Using the above eq., the number of gates affected due to hologram obfuscation for IIR-FIR filter designs is determined to be equal to 6544 gates.

➢ On the other hand the total gate count of both IIR and FIR filter (baseline version) combined is: 5968+3280=9248 gates.

Determination of Gate count affected due to Hologram based Obfuscation of DSP cores

- The total gate count of both IIR and FIR filter (baseline version) combined is: 5968+3280=9248 gates.
- Therefore, the difference in gate count wrt baseline (ΔG_c^b) equal to 9248 – 6544= 2704 gates.
- Since the number of gate present in the hologram obfuscated IIR-FIR filter design is much lesser than the combined baseline equivalent gate count of IIR and FIR filter, thus clear area savings in terms of gate count is achieved.

Analysis on Case Studies

- The following five hologram obfuscated designs produced have been analysed in terms of the security and estimated area / delay overhead (Sengupta and Rathor, 2019):

- 8 point DCT - 4 point IDCT hologram based obfuscated design
- IIR-FIR filter hologram based obfuscated design
- 8 point DIT-FFT - 4 point DIT-FFT hologram based obfuscated design
- 4 point DCT- 4 point IDCT hologram based obfuscated design
- 8 point DCT- 8 point IDCT hologram based obfuscated design

Anirban Sengupta, Mahendra Rathor "Hologram based Structural Obfuscation for DSP Cores", IET Book "Frontiers in Securing Hardware IP Cores: Forensic detective control and obfuscation techniques", 2020, ISBN: 978-1-83953-031-9/978-1-83953-032-6

Strength of Obfuscation Analysis

- The strength of obfuscation of an obfuscated design is defined as:

$$S_{ob} = \frac{G_c^a}{G_c^T}$$

- Where, S_{ob}= Strength of obfuscation.
- G_c^a= Number of gates affected due to obfuscation.
- G_c^T = Total number of gates in a baseline (un-obfuscated) design.

DSP applications	Gate Count of baseline (un-obfuscated design)	Gate Count of Hologram obfuscated design	Gate count Reduction (G_c^a)	Affected gate count
8-pt DCT +4-pt DCT	63936	52352	11584	18.1%
IIR+FIR	9248	6544	2704	29.2 %
8-pt DIT-FFT +4-pt DIT-FFT	14016	9856	4160	29.7%
4-pt DCT+4-pt IDCT	26240	14400	11840	45.1%
8-pt DCT+8-pt IDCT	101632	55424	46208	45.5%

Strength of Obfuscation Analysis

- Comparative study of the strength of obfuscation between hologram obfuscation and HLT obfuscation approach

DSP applications	# of gates affected G_c^a		Affected gate count in % (strength of obfuscation)	
	Hologram Obfuscation	HLT Obfuscation	Hologram Obfuscation	HLT Obfuscation
8-pt DCT +4-pt DCT	11584	2048	18.1%	3.2%
IIR+FIR	2704	224	29.2 %	2.4%
8-pt DIT-FFT +4-pt DIT-FFT	4160	0	29.7%	0%
4-pt DCT+ 4-pt IDCT	11840	512	45.1%	2.0%
8-pt DCT+ 8-pt IDCT	46208	3584	45.5 %	3.5%

Design Delay Analysis

- The additional switching Muxes of a hologram obfuscated design are accountable for extra propagation delay.
- For some hologram obfuscated DSP designs such as IIR-FIR filter and 8 point DCT - 4point DCT, the switching Muxes of size 2:1 are used at input and output of the designs.
- Since all the input Muxes are executed in parallel, therefore delay overhead is estimated as a sum of two 2:1 Muxes (input Mux + output Mux).
- While for some hologram obfuscated DSP designs such as 4 point DCT-IDCT, 8 point DIT-FFT – 4 point DIT-FFT and 8 point DCT-IDCT, the switching Muxes of size 2:1 are used only at the input of the designs.
- Therefore, only the propagation delay of a 2:1 Mux incurs delay overhead for such hologram obfuscated DSP designs .

Conclusion

- We have discussed a structural obfuscation approach based on image hologram feature.
- Like an image hologram, two (or more) DSP applications can be camouflaged into a single design in which each application manifests or evolves on application of correct switching element value.
- This module provides a comprehensive perspective on the hologram based obfuscation of DSP circuits. At the end of this module, the following concepts are delivered:

✓ Necessity of protecting IP cores against piracy and RE threats.
✓ Hologram based structural obfuscation.
✓ Security features of hologram based obfuscation.
✓ Rules of applying hologram based obfuscation for DSP applications.
✓ Design of a hologram obfuscated architecture.
✓ Resemblance of a hologram obfuscated design with an image hologram.
✓ Process of gate count determination of baseline and obfuscated design architectures.
✓ Comparative analysis of several structural obfuscation approaches.

References

- M.C. McFarland, A.C. Parker, R. Camposano (1988), 'Tutorial on high-level synthesis,'*DAC '88 Proceedings of the 25th ACM/IEEE Design Automation,*vol. 27(1), pp. 330-336.
- A. Sengupta (2017), 'Hardware Security of CE Devices [Hardware Matters],'*IEEE Consumer Electronics Mag,* vol. 6(1), pp. 130-133.
- A. Sengupta (2016), 'Intellectual Property Cores: Protection designs for CE products,'*IEEE Consumer Electronics Mag*, vol. 5, no. 1, pp. 83-88.
- A. Sengupta, R. Sedaghat, Z. Zeng (2010), 'A high level synthesis design flow with a novel approach for efficient design space exploration in case of multi-parametric optimization objective,' *Microelectronics Reliability*, vol. 50, Issue: 3, pp. 424-437.
- R. Schneiderman (2010), 'DSPs evolving in consumer electronics applications, '*IEEE Signal Process. Mag*., vol. 27(3), pp.
- R. Torrance and D. James (2009), 'The State-of-the-Art in IC Reverse Engineering,'*in CHES*, vol. 5747, pp. 363-381.
- Y. Lao and K. K. Parhi (2015), "Obfuscating DSP Circuits via High-Level Transformations," *IEEE Trans. on Very Large Scale Integr. (VLSI) Syst*., vol. 23(5), pp. 819–830.
- A. Sengupta, D. Roy, S.P. Mohanty, and P. Corcoran (2017a), 'DSP Design Protection in CE through Algorithmic Transformation based Structural Obfuscation,'*IEEE Transactions on Consumer Electronics*, Vol. 63(4), pp. 467 – 476.
- A. Sengupta and D. Roy (2017), 'Protecting an intellectual property core during architectural synthesis using high-level transformation based obfuscation,'*IET Electronics Letters*, Vol: 53(13), pp. 849 – 851.
- A. Sengupta, S. Bhadauria and S. P. Mohanty (2017b), 'Low-cost security aware HLS methodology,' in *IET Computers & Digital Techniques*, vol. 11 (2), pp. 68-79.
- A. Sengupta, S. Bhadauria, and S. P. Mohanty (2017c), 'TL-HLS: Methodology for Low Cost Hardware Trojan Security Aware Scheduling With Optimal Loop Unrolling Factor During High Level Synthesis,' *IEEE Trans. on CAD of Integrated Circuits and Systems*, vol. 36 (4), pp. 655–668.
- A. Sengupta and M. Rathor (2019), 'Protecting DSP Kernels Using Robust Hologram-Based Obfuscation,' *in IEEE Transactions on Consumer Electronics*, vol. 65(1), pp. 99-108.
- 2016Matrix Technologies, Hologram Features. [Online]. Available: http://www.matrixtechnologies.in/hologram-features.html, last accessed on May 2019.

Thank You